

OBSERVATION TECHNIQUES, ADVANTAGES, DISADVANTAGES AND PERFORMANCE ANALYSIS OF HYBRID CNN-LSTM MODEL FOR SHORT-TERM HOUSEHOLD LOAD FORECASTING

INTRODUCTION

Short-term household electricity load forecasting is an important part of modern smart grid systems. Accurate forecasting helps in power generation planning, energy optimization, demand-side management, and reducing electricity wastage. Individual household load forecasting is more difficult than community-level forecasting because household electricity usage changes continuously depending on user behavior, weather conditions, appliances, and daily activities.

To improve forecasting performance, the paper proposed a hybrid CNN-LSTM deep learning model. The CNN layers extract useful features from time-series electricity data, while LSTM layers learn short-term and long-term temporal dependencies. The proposed model achieved better forecasting accuracy than existing machine learning and LSTM-based models.

OBSERVATION TECHNIQUES

1. Household Energy Consumption Observation

The system continuously observed residential electricity consumption data collected from the Smart Grid Smart City (SGSC) project in Australia. The dataset included half-hourly electricity consumption records from residential customers.

The observations showed that:

- Individual household energy consumption is highly irregular.
- Electricity demand changes during different times of the day.
- Different households exhibit different energy usage patterns.

The system analyzed energy consumption behavior over 92 days for 69 customers.

2. Clustering Observation

The DBSCAN clustering technique was used to observe regularity and irregularity in household electricity consumption profiles.

The clustering observation identified:

- Major clusters

- Minor clusters
- Outliers

Observations revealed:

- Some households showed regular electricity patterns.
- Some households showed highly irregular patterns with many outliers.
- Prediction becomes difficult when outliers increase.

The clustering method helped classify customers according to energy usage behavior.

3. Load Profile Observation

The system observed daily electricity load profiles for individual households and aggregated community loads.

Key observations:

- Individual household load profiles fluctuate sharply.
- Aggregated load profiles are smoother and easier to predict.
- Peak electricity demand usually occurs during evening hours.

This observation confirmed that individual household forecasting is more challenging than community-level forecasting.

4. Feature Extraction Observation

The CNN layer was used for automatic feature extraction from electricity time-series data.

The input features included:

- Energy consumption sequence
- Hour of the day
- Day of the week
- Holiday indicator

The CNN layer observed:

- Local consumption patterns
- Short-term fluctuations
- Important hidden features

MaxPooling and ReLU layers improved feature learning and reduced computational complexity.

5. Sequence Learning Observation

LSTM layers were used to observe temporal dependencies in electricity usage sequences.

The LSTM layers observed:

- Short-term dependencies
- Long-term dependencies
- Sequential behavior patterns

The model used three LSTM layers with 20 hidden nodes each.

This observation improved forecasting performance for both single-step and multi-step prediction.

6. Training Observation

The model training process continuously observed:

- Mean Absolute Error (MAE)
- Validation loss
- Epoch performance
- Weight optimization

The dataset was divided into:

- 70% Training Data
- 20% Validation Data
- 10% Test Data

The Adam optimizer was used during training.

Dropout layers were added to reduce overfitting.

7. Single-Step Forecast Observation

The proposed CNN-LSTM model was compared with:

- LSTM

- KNN
- BPNN
- ELM
- IS-HF

The model achieved:

- Average MAPE = 40.38%

while the LSTM model achieved:

- Average MAPE = 44.06%

The proposed model performed better for 57 out of 69 households.

8. Multi-Step Forecast Observation

The system also observed forecasting performance for:

- 1-step prediction
- 2-step prediction
- 6-step prediction

The proposed CNN-LSTM model consistently outperformed the standard LSTM model.

Observed improvements:

- 4.01% improvement for 1-step forecasting
- 4.76% improvement for 2-step forecasting
- 5.98% improvement for 6-step forecasting

9. Customer Behavior Observation

The system observed electricity usage behavior during:

- Breakfast period
- Daytime period
- Evening period
- Overnight period

Different households showed completely different energy usage behavior patterns.

ADVANTAGES

1. Improved Prediction Accuracy

The hybrid CNN-LSTM model achieved better prediction accuracy compared to traditional machine learning and LSTM-only models.

2. Efficient Feature Extraction

CNN layers automatically extracted important features from time-series electricity data.

3. Better Temporal Learning

LSTM layers captured both short-term and long-term dependencies effectively.

4. Reduced Prediction Error

The model achieved lower Mean Absolute Percentage Error (MAPE) values.

5. Handles Complex Household Patterns

The hybrid model successfully handled highly irregular household electricity usage behavior.

6. Better Multi-Step Forecasting

The proposed model achieved better forecasting performance for future time horizons up to 3 hours ahead.

7. Noise Reduction

CNN layers helped filter noisy input data and improve forecasting stability.

8. Smart Grid Support

The model supports:

- Smart grid systems
- Demand-side management

- Energy optimization
- Load balancing

9. Real-Time Forecasting Capability

The framework can be integrated into real-time smart energy monitoring systems.

10. Scalability

The model can be extended for:

- Large residential datasets
- Community-level forecasting
- Smart city applications

DISADVANTAGES

1. High Computational Complexity

The hybrid CNN-LSTM architecture requires significant computational resources.

2. Large Training Time

Deep learning models require long training durations because multiple CNN and LSTM layers are involved.

3. Dependency on Data Quality

Prediction performance decreases if the dataset contains:

- Missing values
- Noisy records
- Incomplete data

4. Difficult Hyperparameter Tuning

The model requires careful tuning of:

- Learning rate
- Epochs
- Kernel size
- Hidden layers
- Batch size

5. High Memory Consumption

Training deep learning models requires large RAM and GPU memory.

6. Overfitting Risk

The model may overfit if trained excessively on limited data.

7. Complex Architecture

The CNN-LSTM hybrid structure is more difficult to implement than traditional forecasting models.

8. Difficult Interpretability

Deep learning systems behave like black-box models and are harder to interpret.

9. Outlier Sensitivity

Prediction accuracy decreases significantly when household electricity profiles contain many outliers.

10. Hardware Cost

Real-time deployment may require:

- High-performance servers
- GPU hardware
- Cloud infrastructure

PERFORMANCE ANALYSIS

1. CNN-LSTM Performance

The proposed CNN-LSTM model achieved the best overall forecasting performance.

Results:

- Average MAPE = 40.38%
- Better than all compared methods

The model effectively captured temporal patterns and reduced forecasting errors.

2. LSTM Performance

The LSTM-only model achieved:

- Average MAPE = 44.06%

Although LSTM captured sequence information effectively, it lacked strong feature extraction capability compared to CNN-LSTM.

3. KNN Performance

KNN showed:

- High prediction error
- Poor scalability
- Weak forecasting performance

4. BPNN Performance

Backpropagation Neural Network (BPNN) achieved moderate forecasting performance but struggled with complex nonlinear patterns.

5. ELM Performance

Extreme Learning Machine (ELM) showed lower prediction accuracy and poor generalization capability.

6. Clustering Performance

Clustering analysis helped group households with similar electricity usage patterns.

The clustering-based forecasting approach improved prediction performance for some households but not all.

7. Outlier Impact Analysis

The observations revealed:

- More outliers lead to larger forecasting errors.
- Regular household patterns are easier to predict.
- Highly irregular households produce high MAPE values.

CONCLUSION

The hybrid CNN-LSTM deep learning framework provides an efficient solution for short-term individual household load forecasting. Observation techniques such as clustering analysis, load profile analysis, feature extraction, and sequence learning significantly improved prediction performance.

The proposed model outperformed existing methods including LSTM, KNN, BPNN, and ELM. The CNN layers extracted important features while the LSTM layers captured temporal dependencies effectively.

Although the system introduces challenges such as computational complexity, high training time, and hardware requirements, it remains a highly effective forecasting framework for future smart grid and residential energy management applications.